Faculty of Engineering & Technology

Department of Electrical and Computer Engineering

ADVANCED DIGITAL DESIGN - ENCS3310

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Smart Traffic Light Controller with Pedestrian Request

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State Encoding

The traffic light controller is implemented as a Moore finite state machine (FSM) with 3-bit binary encoding, allowing up to 8 distinct states. I chose this encoding for simplicity and efficient hardware utilization.

State Name	Binary Encoding	Description
S_MG_MIN	000	Main road green (minimum guaranteed time)
S_MG_EXT	001	Main road green (extended while no requests)
S_MY	010	Main road yellow
S_AR1	011	All-red before side road
S_SG	100	Side road green
S_SY	101	Side road yellow
S_AR2	110	All-red before pedestrian/main
S_PG	111	Pedestrian walk green

Output Encoding

Traffic light colors are encoded as:

- RED = 3'b100
- YELLOW = 3'b010
- GREEN = 3'b001

Outputs depend only on the current state (Moore):

- Main road green during S_MG_MIN and S_MG_EXT
- Side road green during S_SG
- Pedestrian walk asserted during S_PG

Timing Behavior

All state durations are controlled using a countdown timer, loaded upon state entry and decremented every clock cycle. Timing parameters are scalable using T_SCALE.

Parameter	Value (seconds)	Description
MAIN_GREEN_T	60	Minimum main road green time
SIDE_GREEN_T	40	Fixed side road green time
PED_GREEN_T	40	Fixed pedestrians walk time
YELLOW_T	5	Yellow light duration
ALL_RED_T	1	All-red clearance time

Main Road Green: Guaranteed minimum duration (S_MG_MIN) Automatically extends (S_MG_EXT) if no vehicle (VS) or pedestrian (PB) requests are pending

Side Road and Pedestrian: Exact fixed durations (40 seconds)

Yellow and All-Red: Used as safety transitions between conflicting green phases

Pending Requests: Vehicle and pedestrian requests are latched and serviced in the next appropriate cycle Pedestrian requests during side green are queued for the next cycle

Results and Verification

A self-checking testbench was developed to compare the DUT (traffic_ctrl) against a behavioral reference model (traffic_ctrl_ref) on a cycle-by-cycle basis. Outputs were sampled on every negative clock edge after reset.

The following signals were compared:

1- light_main
2- light_side
3- walk

Any mismatch increments an error counter and prints diagnostic information.

Simulation Results

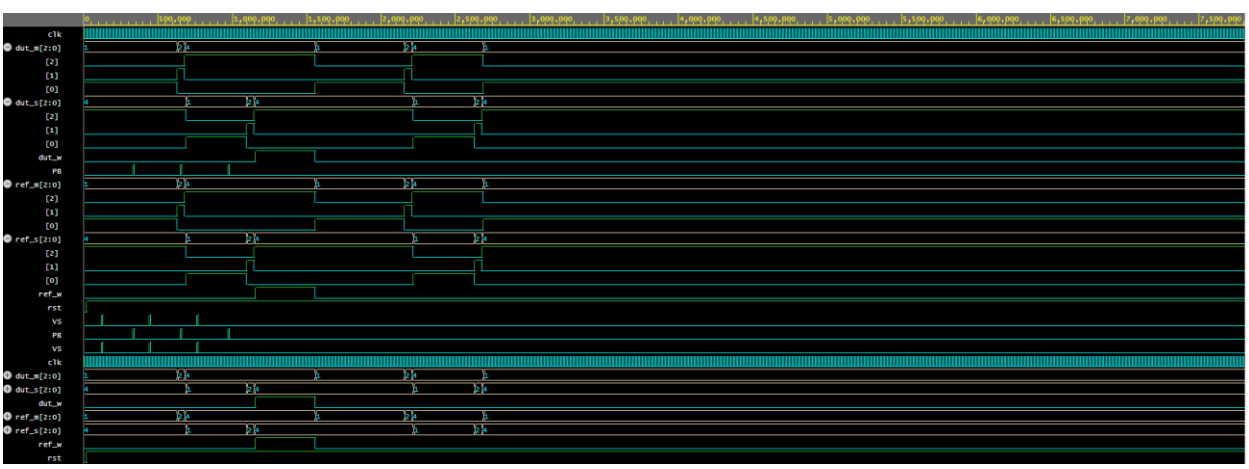
- Multiple vehicle sensor (VS) and pedestrian button (PB) activations were applied
- Full traffic cycles were observed, including:
    - Main → Side → Pedestrian → Main transitions
    - Extended main green behavior
    - Queued pedestrian requests
- No mismatches were detected

Waveform Analysis

The waveform confirms:

- Correct state sequencing
- Proper enforcement of timing constraints
- Identical outputs between DUT and reference model
- Clean reset behavior and stable outputs

This demonstrates that the RTL implementation is functionally correct and fully compliant with the given specification



The simulation waveform provides detailed confirmation of the correct functional behavior of the traffic light controller. After the asynchronous reset is asserted and then released, the controller initializes in the Main Road Green minimum state (S_MG_MIN). At this point, the

waveform shows light_main = 3'b001 (GREEN), light_side = 3'b100 (RED), and walk = 0, indicating correct default operation. The internal countdown timer is loaded with the programmed minimum main green duration and decrements on every rising edge of the clock. During this phase, if no vehicle (VS) or pedestrian (PB) requests are present, the controller transitions into the extended main green state (S_MG_EXT), as observed in the waveform by the continued assertion of light_main = GREEN while the timer remains at zero. When a vehicle sensor request (VS) is asserted, the waveform shows that the request is latched using the veh_pending signal and serviced only after the minimum green time has elapsed. Upon timer expiration, the FSM transitions synchronously on the rising clock edge to the Main Yellow state (S_MY), where light_main changes from GREEN to YELLOW for the programmed yellow duration. This is followed by an All Red clearance state (S_AR1), during which both main and side road lights remain RED. After the all red interval, the controller transitions to the Side Road Green state (S_SG). The waveform confirms that light_side = GREEN for the full fixed duration while light_main remains RED. During this phase, a pedestrian button press (PB) is asserted and correctly latched as a pending request, without interrupting the ongoing side road green phase. Once the side road timer expires, the FSM transitions through Side Yellow (S_SY) and a second All Red state (S_AR2). Since a pedestrian request is pending, the controller then enters the Pedestrian Walk state (S_PG). The waveform clearly shows walk = 1 asserted only during this state for the programmed pedestrian duration, confirming correct Moore FSM output behavior. At the completion of the pedestrian phase, the FSM transitions back to Main Road Green (S_MG_MIN), restoring the default traffic flow. Throughout the simulation, all state transitions occur synchronously on clock edges, timing constraints are strictly enforced by the countdown timer, and no output glitches are observed.

Finally, a direct comparison of the DUT outputs (dut_m, dut_s, dut_w) and reference model outputs (ref_m, ref_s, ref_w) in the waveform confirms that they match exactly at every cycle, validating the correctness of the FSM implementation, timing control, and request handling logic.